

Experiment-5

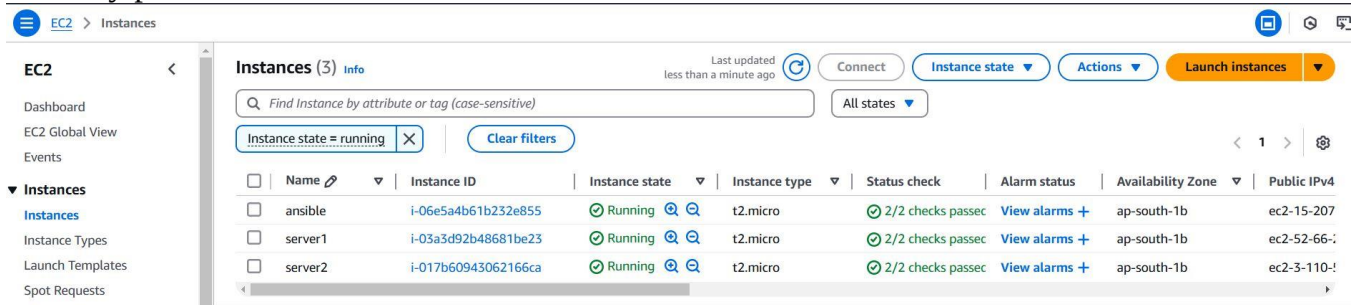
Aim: Ansible Setup and SSH key

(Note: All Blue Words is a command, just copy and paste)

Require Software & Tools: AWS Account, Puttygen software, Wincp Software.

Procedure:

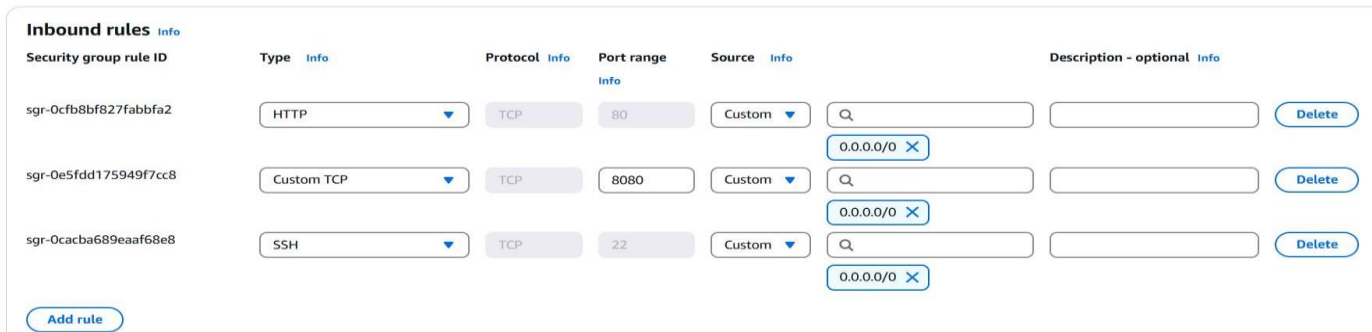
Step-1: Login AWS and Create three Instances on AWS account using ubuntu OS and save key pair also.



Step-2: change the 2nd and 3rd instance name as a server1 and server2 (you can named anything) -> Edit inbound rules of 1st instance (ansible). Edit inbound rules like given details.

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.



Step-3: Now come on instance and select 1st instance (ansible) and click on connect. It will open a linux terminal -> type **Clear** command.

```
Usage of /: 24.9% of 6.71GB  Users logged in: 0
Memory usage: 20%          IPv4 address for enX0: 172.31.13.107
Swap usage: 0%
```

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.
To check for new updates run: `sudo apt update`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "`sudo <command>`".
See "`man sudo_root`" for details.

ubuntu@ip-172-31-13-107:~\$

i-06e5a4b61b232e855 (ansible)

PublicIPs: 15.207.222.240 PrivateIPs: 172.31.13.107

Step-4: type command `sudo apt update` and hit enter.

Step-5: type command `sudo apt install ansible` and hit enter -> it will ask yes/no type yes and hit enter.

Step-6: type command `sudo apt install openssh-client` and hit enter.

Step-7: type command `sudo apt install openssh-server` and hit enter.

Step-8: type command `cd ~/.ssh/` and hit enter.

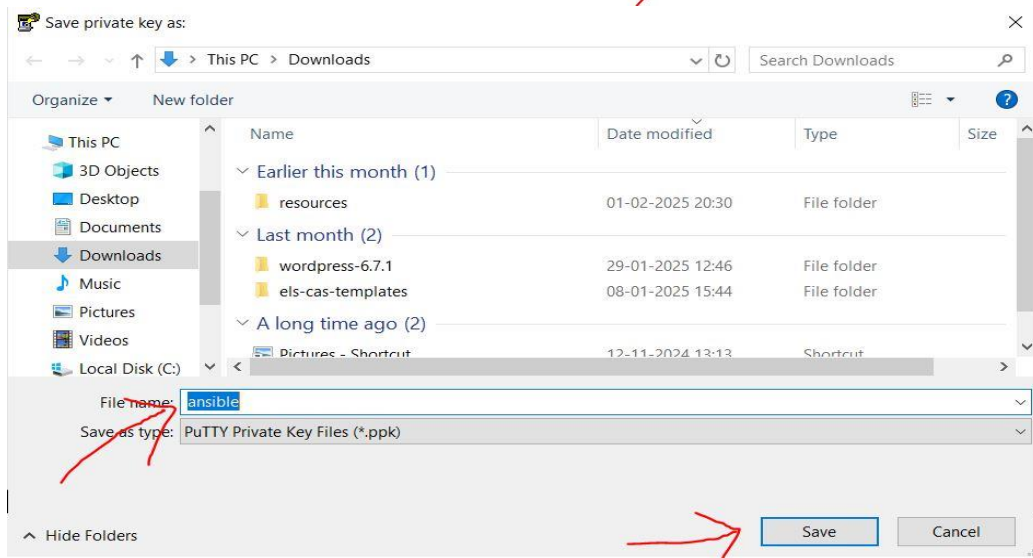
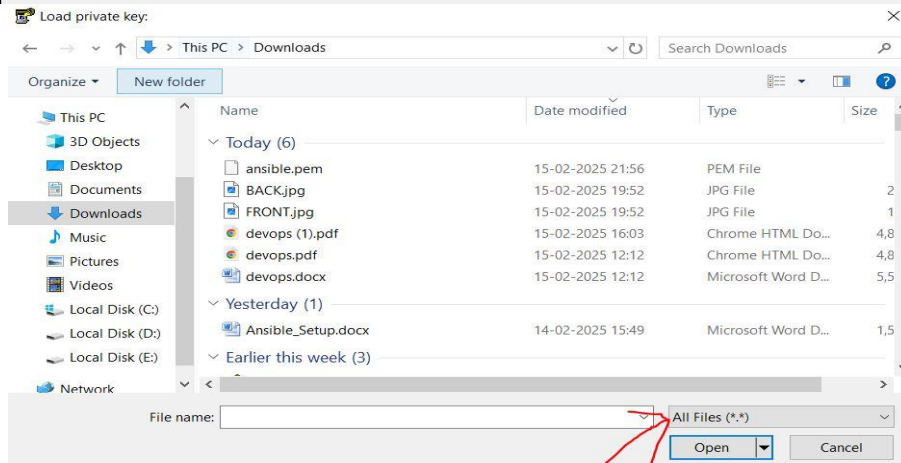
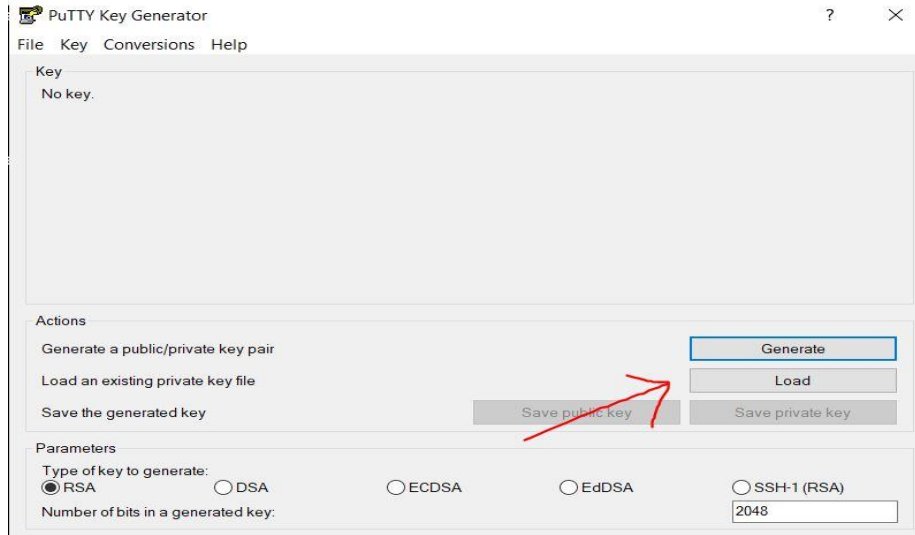
Step-9: type command `ls` and hit enter.

Step-10: type command `cd` and hit enter.

Step-11: type command `cd -` and hit enter. (it will show ssh path)

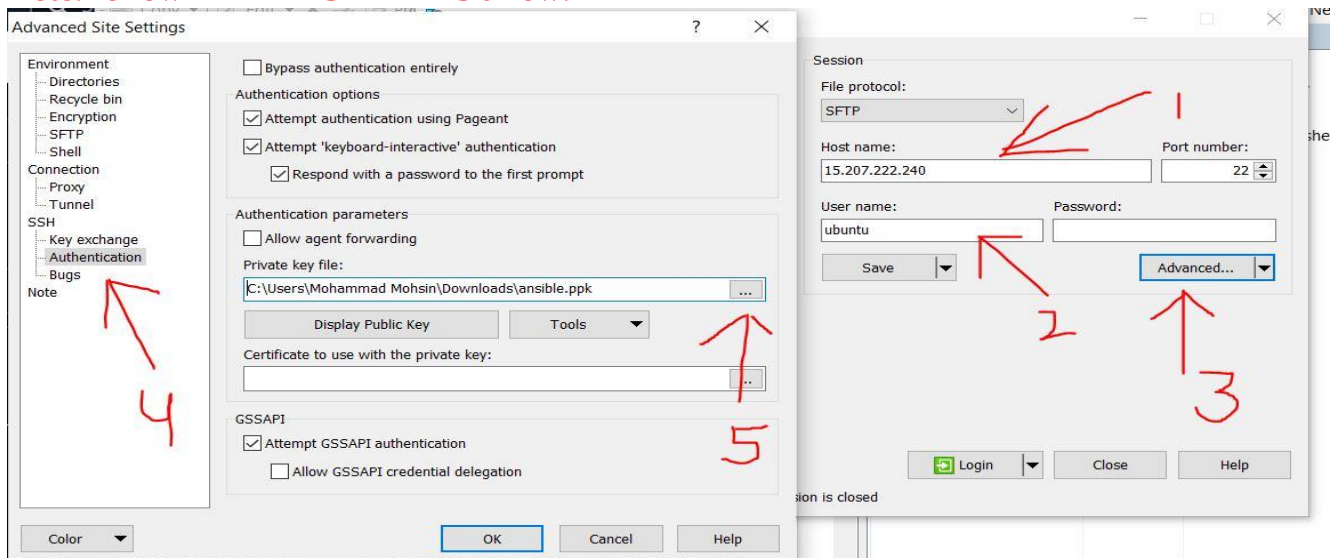
Step-12: now again type command `cd` and hit enter.

Step-13: open **puttygen software** -> click on **load** -> select as **all files** -> select **ansible.pem** (your key pair) -> click on **save** and -> click on **save private key** and save the file with **.ppk extension**.



Step-14: Now open **winCP** software -> it will ask for **host name** just copy and paste your **ansibleinstance Ip**-> In the place of **user name** type **Ubuntu** -> click on **advance** -> select

authentication -> in the place of **private key** select your downloaded .ppk file - > click on **Ok** -> click on **Login**. (it will connect to your server and show a **Ubuntu wincp window**)
Note: follow 1 -> 2 -> 3 -> 4 -> 5 arrow.



Step-15: after connecting to your server it will show **two directory** (one is your local disk, another is ssh directory) -> right click in **ssh directory** -> click on **new** -> select **directory** -> create a **folder** name as **key** -> now select your .pemfile from left side local disk directory and **drag in your key folder**.

Step-16: Now come in your terminal -> type **cd** command and hit enter -> type **ls** command hit enter, you will see key.

```
ubuntu@ip-172-31-13-107:~/.ssh$ cd
ubuntu@ip-172-31-13-107:~$ ls
ubuntu@ip-172-31-13-107:~$ cd
ubuntu@ip-172-31-13-107:~$ ls
key
ubuntu@ip-172-31-13-107:~$
```

Step-17: Now type **cd key/** and hit enter, you will see ansible.pem

Step-18: Now type **cp ansible.pem ~/.ssh/** command and hit enter -> after that type **cd** hit enter -> after that **ls ~/.ssh/** and hit enter, you will see ansible.pem and authorized_key

Step-19: Now create one more directory using **mkdir ansible-script** command and hit enter -> select directory using **cd ansible-script/** command and hit enter.

Step-20: Now write **vi inventory.ini** command and hit enter -> it will open your Vim Editor -> copy the given command and paste in your Vim editor -> after that in the place of [server1-ip] copy and paste server1 instance public ip and in the place of [server2-ip] copy and paste server2 instance public ip -> after that save this vim editor using **:wq!** Command and hit enter

Command:

```
[webservers]
server1 ansible_host=34.230.5.214 ansible_user=ubuntu ansible_ssh_private_key_file=/home/ubuntu/.ssh/ansible.pem
server2 ansible_host=98.93.243.50 ansible_user=ubuntu ansible_ssh_private_key_file=/home/ubuntu/.ssh/ansible.pem
```

Step-21: Now write the command `vi playbook.yml` and create a yml file -> it will again open your Vim editor -> copy and paste the given command in your vim editor after save and exit using command `:wq!` and hit enter.

Command:

```
---
- name: Install Nginx on AWS EC2
  hosts: webservers
  become: yes

  tasks:
    - name: Install Nginx
      apt:
        name: nginx
        state: present
        update_cache: yes
        when: ansible_os_family == "Debian"

    - name: Start and enable nginx server
      systemd:
        name: nginx
        state: started
        enabled: yes
```

Step-22: Now you can check your .ini file and .yml file using `cat inventory.ini` command and `cat playbook.yml` command.

Step-23: Now type `ansible -i inventory.ini webservers -m ping` command and hit enter. Its showing server 1 and 2 success means no issue. If its show server1 and 2 unsuccess or unreachable means follow below commands.

```
chmod 400 /home/ubuntu/.ssh/ansible.pem
```

```
ls -l /home/ubuntu/.ssh/ansible.pem
```

Step-24: Now type `ansible-playbook -i inventory.ini playbook.yml` command and hit enter -> it will ask for type `yes` and again `yes` and again `yes` .It will run your server properly. After that you can check your server in any browser using server ip.

OUTPUT:-

```
ubuntu@ip-172-31-4-180:~$ sudo apt-get install -y ansible
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
ubuntu@ip-172-31-4-180:~$ ansible-playbook playbook1.yml --syntax-check
playbook: playbook1.yml
ubuntu@ip-172-31-4-180:~$ vim playbook1.yml
ubuntu@ip-172-31-4-180:~$ ansible-playbook playbook1.yml --syntax-check
playbook: playbook1.yml
ubuntu@ip-172-31-4-180:~$ ansible-playbook playbook1.yml -b

PLAY [Configure tomcat9] *****
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.7.139]
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.5.200]

TASK [Install tomcat9] *****
ok: [172.31.7.139]
ok: [172.31.5.200]

TASK [Change port no of tomcat 8080 to 9090] *****
ok: [172.31.7.139]
ok: [172.31.5.200]

PLAY RECAP *****
172.31.5.200      : ok=3  changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.7.139     : ok=3  changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

⚠ Not secure 43.204.218.193

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

⚠ Not secure 13.233.103.243

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.